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Lab/Project Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Problem understanding & Analysis	7					
Implementation	8					
Report writing	10					
Total obtained mark						
Comments						

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Capstone Project

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PROJECT PROPOSAL REPORT

Statute-Grounded Bangla Legal Assistant

A Retrieval-Augmented Generation (RAG) system that answers legal questions in Bangla and cites the exact statute and section

Course	SE331
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1. Introduction

1.1 Background Overview

Application domain. The application domain of this project is **Legal Technology (LegalTech)**, specifically legal information access and public legal awareness in Bangladesh. In Bangladesh, laws are written as Acts, and each Act is divided into numbered sections. Official consolidated texts of the Acts are published online by the Legislative and Parliamentary Affairs Division through the Bangladesh Laws portal (bdlaws.minlaw.gov.bd). Many Acts are available in Bangla, and older laws such as the Penal Code, 1860 are mainly available in English.

Current practice. When an ordinary person wants to know what the law says about a situation (for example, a worker whose employer has not paid wages, or a consumer sold an expired product), the usual options are: asking a lawyer, asking friends or relatives, searching Facebook groups or general web pages, or reading the Act directly. A lawyer's time costs money and not everybody can afford it. Reading an Act directly is difficult because legal Bangla is formal, one situation may involve several sections spread across different Acts, and the portal supports only plain keyword search, which fails when the user describes the problem in everyday words instead of legal terms.

Use of AI in this domain. General-purpose chatbots (such as ChatGPT or Gemini) can answer legal questions in Bangla, but they are not tied to the actual text of Bangladeshi statutes. They may quote a section number that does not exist, mix Bangladeshi law with Indian or Pakistani law, or use an outdated or repealed version, and they usually cannot show the exact legal text behind an answer. A few local legal-information websites exist, but they are mostly static article collections. Therefore, a Bangla-first assistant that is grounded in the official statute text and always shows its source is still largely missing.

1.2 Problem Statement

Problem. Ordinary citizens cannot easily find which law and which section applies to their everyday situation, because (a) statutes are long and written in formal legal language, (b) keyword search on the official portal does not understand questions written in everyday Bangla, and (c) general AI chatbots give fluent answers that are not verifiably based on Bangladeshi statutes and may be wrong.

Gaps in existing systems.

- The official portal supports lookup by Act name and keyword but cannot answer a natural-language question.
- General chatbots can hallucinate section numbers or content and do not reveal their source.
- Most existing tools are English-first; Bangla questions are handled less reliably.
- Existing sources rarely distinguish between 'this is what the statute says' and 'this is advice'.

Who is affected. Members of the general public (workers, consumers, women facing domestic problems, internet users facing online harassment), law students who need to locate provisions quickly, and community organisations or legal-aid volunteers who answer basic legal queries repeatedly.

Why software is needed. A software system can be available at any time, can search thousands of sections in seconds, and can present the original legal text next to a simple explanation, which is not practical to do manually for every person.

Why AI is needed. Today, a person must manually read through Acts, decide which section may be relevant, and interpret the formal language. AI can automate exactly these steps: (1) *semantic search* matches a question written in everyday Bangla to the relevant sections even when no words are shared; and (2) a *language model* rewrites the retrieved legal text into a short, plain-Bangla explanation. Rule-based or keyword approaches cannot do this reliably because the same situation can be described in many different ways.

Scope realism. The project is intentionally limited to a fixed set of selected Acts and to *legal information*. It does not claim to give legal advice, predict court outcomes, or cover case law.

1.3 Objectives

General Objective. To develop a Bangla-language legal information assistant that helps citizens understand which provisions of selected Bangladeshi statutes relate to their situation, by answering questions in simple Bangla with exact Act and section references.

AI Objective. To build a Retrieval-Augmented Generation (RAG) pipeline that (i) retrieves the most relevant statute sections for a Bangla question using multilingual semantic search, (ii) generates an answer that is based only on the retrieved sections and cites them, and (iii) declines to answer when the knowledge base does not contain sufficient support. The pipeline will be evaluated on a self-prepared Bangla question set for retrieval accuracy and answer faithfulness.

1.4 Scope

In scope

- A knowledge base built from the official text of a limited set of Acts (initially four to five; proposed: Bangladesh Labour Act 2006, Consumer Rights Protection Act 2009, Cyber Security Act 2023, Domestic Violence (Prevention and Protection) Act 2010, and selected chapters of the Penal Code 1860). The final list will be fixed in the first two weeks according to text availability and workload.
- Question input in Bangla (typed text; Banglish/English questions handled on a best-effort basis).
- Retrieval of relevant sections, and a plain-Bangla answer with citations (Act name, section number) and the original section text.
- A 'not enough information' response when retrieval confidence is low.
- A simple web interface, an answer feedback button (helpful / not helpful), and a mandatory legal-information disclaimer.
- An evaluation module with a test set of at least 100 questions.

Out of scope

- Legal advice, drafting legal documents, or predicting the outcome of a case.
- Court judgments, case law, rules and regulations, gazette notifications, and all Acts not in the selected list.
- Voice input, OCR of scanned documents, and mobile app development.
- Automatically tracking amendments in real time (the knowledge base is a fixed snapshot, with its date shown).
- Handling user accounts with personal case files; the system does not store personal case details.

What the AI will do: understand the question, find relevant sections, summarise them in simple Bangla, cite them, and refuse when unsupported. **What the AI will not do:** decide who is legally right or wrong, recommend legal action, or answer from knowledge outside the retrieved statute text.

System boundary. The system takes a text question as input and returns an answer based only on the stored statute snapshot. Anything requiring a lawyer's judgement is outside the boundary and the user is directed to a qualified lawyer or legal-aid service.

1.5 Stakeholders

Stakeholder	Role	Interest in the system
General citizens (end users)	Ask questions and read answers	Quick, understandable, trustworthy information in Bangla at no cost
Law students and teachers	Use as a lookup and learning tool	Fast navigation to correct sections with exact text
Legal-aid volunteers / NGOs	Use for first-level information	Reduce repetitive basic queries; verify source text easily
Lawyers (advisors / reviewers)	Review sample answers (optional)	Ensure information is not misleading
System administrator (project team)	Maintain knowledge base, models, and hosting	Reliable operation, updating statute data, monitoring feedback
Course instructor / university	Evaluate the project	Feasible scope, sound AI design, proper evaluation
Legislative authority (data source)	Publisher of official statute text	Correct and fair use of the official text with attribution

1.6 Proposed Solution

The proposed solution is a web-based assistant built on a RAG pipeline. Instead of letting a language model answer from its own memory, the system first retrieves the relevant sections from the official statute text and then asks the model to answer using only those sections.

Workflow

1. **Preparation (offline).** Collect the official text of the selected Acts, clean it, and split it section by section, keeping metadata (Act, chapter, section number, title). Convert each section into an embedding vector and store it in a vector database. [AI: **embedding model**]
2. **Question.** The user types a question in Bangla.
3. **Retrieval.** The question is embedded and the most similar sections are retrieved (top-k), then optionally re-ranked. [AI: **semantic search**]
4. **Generation.** A language model receives the question and only the retrieved sections, and writes a short plain-Bangla answer with citations. [AI: **LLM**]
5. **Verification.** The system checks that every cited section actually exists in the retrieved set. If retrieval confidence is low or the check fails, it replies that it cannot find a supported answer. [AI-assisted check]
6. **Display.** The answer, the cited Acts and sections, the original text, and a disclaimer are shown. The user can give feedback.

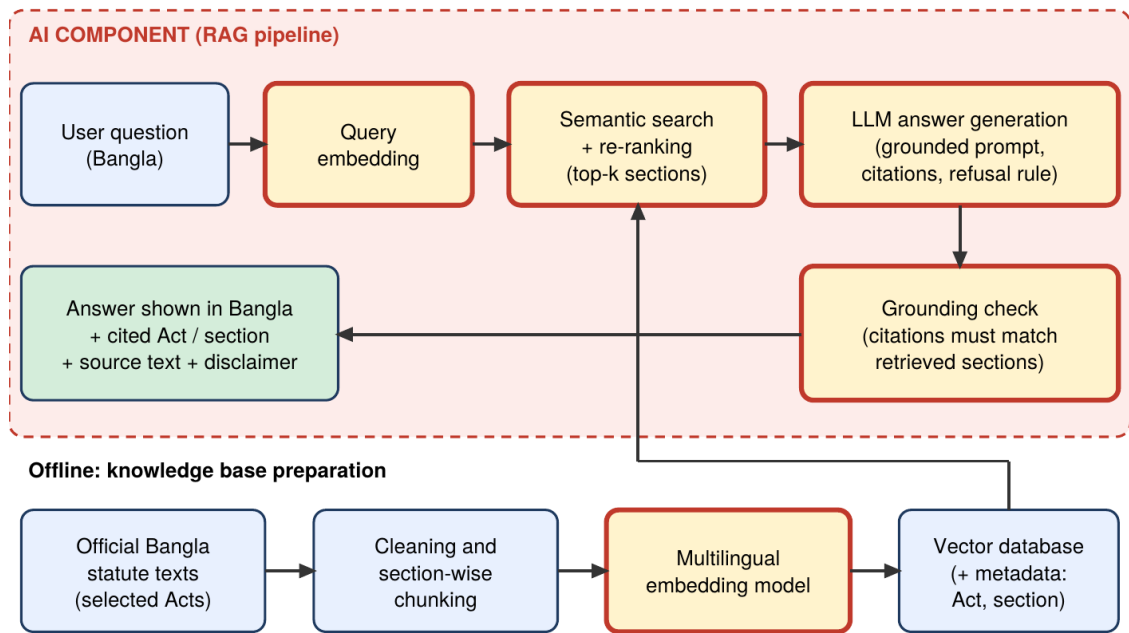


Figure 1: System overview. Components marked in red and yellow are AI components.

1.7 AI Features

Existing AI features and their limitations

Existing feature	Limitation
General LLM chatbots (ChatGPT, Gemini, etc.) answer legal questions in many languages	Not grounded in Bangladeshi statutes; may invent sections or mix jurisdictions; no verifiable source; may be outdated
Keyword search on the official laws portal	Needs the user to know the Act and legal terms; no understanding of everyday Bangla; no explanation
Generic English legal-research tools	Cover other countries' laws; English-first; mostly paid

Proposed AI features

Feature	Input	Output	Manual work replaced
F1. Semantic statute search	Bangla question (everyday wording)	Ranked list of relevant sections with scores	Reading Acts and guessing keywords
F2. Grounded answer generation	Question + retrieved sections	Plain-Bangla explanation with Act and section citations	Interpreting formal legal language
F3. Unsupported-question detection	Retrieval scores + generated answer	'Cannot answer from available statutes' message, or the answer	A person judging if the law actually covers the question
F4. Citation verification	Answer citations + retrieved sections	Pass/fail flag; answer blocked if citations are invalid	Manually checking each cited section

AI techniques. Retrieval-Augmented Generation (RAG) over our own corpus of statute text; multilingual / Bangla-capable text embeddings with a vector database (semantic search); optional cross-encoder re-ranking; LLM integration through an API with a constrained prompt; and Bangla language AI (Bangla question understanding and Bangla answer generation). Evaluation will use retrieval metrics (Hit@k, MRR) and answer faithfulness checks.

Why AI is essential. If the AI components are removed, the system cannot match an everyday Bangla question to the correct legal sections or explain them in simple language, which is its main purpose; only a plain list of Acts would remain, equivalent to the current portal. The semantic retrieval and the grounded generation are the core of the product, not an add-on.

2. System Requirements

2.1 Functional Requirements

ID	Requirement
FR1	The system shall accept a question typed in Bangla through a web interface.
FR2	The system shall retrieve the top-k (default k = 5) most relevant statute sections for a question.
FR3	The system shall generate an answer in simple Bangla using only the retrieved sections.
FR4	The system shall display, for every answer, the Act name, section number, and original section text it is based on.
FR5	The system shall display a 'cannot find a supported answer' message when the best retrieval score is below a configured threshold.
FR6	The system shall verify that every section cited in an answer is among the retrieved sections, and shall block answers that fail the check.
FR7	The system shall show a legal-information disclaimer with every answer.
FR8	The system shall allow the user to filter the search by Act (optional).
FR9	The system shall allow the user to mark an answer as helpful or not helpful and optionally add a comment.
FR10	The system shall store questions and feedback without personal identifiers, for evaluation only.
FR11	An administrator shall be able to add or update the statute data and rebuild the index using a script.
FR12	The system shall display the date of the statute snapshot used.
FR13	The system shall include an evaluation script that reports Hit@k, MRR and faithfulness on the test question set.

2.2 AI Requirements

ID	Requirement
AIR1	The retrieval component shall return a correct relevant section within the top 5 results for at least 75% of questions in the test set (target; to be reported honestly after evaluation).
AIR2	At least 90% of the section citations in generated answers shall refer to sections that exist in the retrieved context (faithfulness target).
AIR3	The system shall refuse, or state insufficient information, for questions in the test set that are out of scope (for example, questions about unrelated topics or un-indexed Acts) in at least 80% of cases.
AIR4	The generated answers shall be in Bangla and not exceed about 200 words, excluding the quoted section text.

ID	Requirement
AIR5	The system shall not give instructions such as 'file a case against X' or predict outcomes; the prompt and an output check shall enforce a neutral, informational style.

2.3 Non-Functional Requirements

- **Performance:** a typical answer shall be shown within 10 seconds on a normal internet connection.
- **Usability:** the interface shall be simple, support Bangla fonts correctly, and be usable on mobile browsers.
- **Privacy:** no names, phone numbers or account data shall be collected; users are warned not to enter personal details.
- **Maintainability:** data preparation, indexing and evaluation shall be separate scripts so that statute data can be refreshed.
- **Transparency:** the sources and the statute snapshot date shall always be visible.

3. Tools and Technologies

Category	Tools (proposed)
Programming languages	Python (backend, AI pipeline, data processing); JavaScript/HTML/CSS (frontend)
Backend framework	FastAPI
Frontend	Streamlit for the prototype, or React for a custom interface (final choice by team skill)
Data collection and processing	Official Bangladesh Laws portal as source; Requests/BeautifulSoup for collection (respecting terms of use), PDF text extraction libraries where needed, regular expressions and Pandas for cleaning and section-wise splitting
Databases	ChromaDB or FAISS (vector store); SQLite (feedback and logs)
Embedding models	Open multilingual embedding model with Bangla support (e.g., multilingual-e5 or BGE-M3 via Sentence-Transformers); the best one will be selected by testing on our question set
Language model	A Bangla-capable LLM through an API (e.g., Claude API or another provider); prompt-constrained to retrieved context
Re-ranking (optional)	Multilingual cross-encoder re-ranker
RAG orchestration	LangChain or LlamaIndex, or a small custom pipeline
Evaluation	Own test set (CSV); Hit@k, MRR; manual faithfulness check plus a model-based judge as a supplement
Version control and collaboration	Git and GitHub; Google Drive for documents
Deployment (demo)	Local machine or a free-tier cloud service (e.g., Hugging Face Spaces or Render)

4. Project Timeline and Work Plan

The plan assumes a 14-week semester and a team of four members (Member 1 to Member 4); names and durations will be adjusted to the actual team and course calendar.

4.1 Milestone Table

Week	Milestone	Main tasks	Deliverable
1-2	M1: Planning and data selection	Finalise list of Acts; study text availability; set up repository; define question-set format	Proposal, repository, Act list
3-4	M2: Knowledge base ready	Collect and clean statute text; split by section; add metadata	Clean section-wise dataset
5-6	M3: Retrieval prototype	Compare embedding models; build vector index; test retrieval	Working semantic search
7-8	M4: RAG answer generation	Design prompts; integrate LLM; implement citations, refusal and verification	End-to-end backend
9-10	M5: Interface and integration	Build web UI; connect backend; feedback and disclaimer	Working demo
11-12	M6: Evaluation and improvement	Finalise 100+ question test set; run metrics; error analysis; tune threshold and prompts	Evaluation results
13	M7: Testing and documentation	User testing with classmates; bug fixing; final report	Final report draft
14	M8: Final submission and demo	Final presentation and demo; code cleanup	Final report, demo, source code

4.2 Task Distribution Among Team Members

Member	Responsibilities
Member 1 (Data and knowledge base)	Statute collection, cleaning, section splitting, metadata, snapshot documentation
Member 2 (Retrieval)	Embedding model comparison, vector database, re-ranking, retrieval evaluation
Member 3 (Generation and verification)	Prompt design, LLM integration, refusal logic, citation verification, safety checks
Member 4 (Interface and evaluation)	Frontend, API integration, feedback storage, test question set, user testing
All members	Documentation, weekly meetings, code review, presentation

5. Additional Sections

5.1 Existing System

Currently, people use (a) the official laws portal for browsing and keyword search, (b) general web search and social-media groups, and (c) general chatbots. These are free and available, but they do not combine Bangla question understanding, exact statute citations, and verification. Our system differs by restricting answers to a verified statute snapshot and showing the source text beside the explanation.

5.2 System Architecture

The system has four layers: **(1) Data layer** - cleaned statute sections and metadata, a vector index, and a small database for feedback; **(2) AI layer** - embedding model, retriever and re-ranker, LLM generator, and citation verifier; **(3) Application layer** - FastAPI services that orchestrate the pipeline; **(4) Presentation layer** - a web interface that shows the answer, sources and disclaimer. Figure 1 shows the flow and marks the AI components.

5.3 Ethical, Legal, and Social Considerations

- **Not legal advice:** every answer carries a disclaimer and a suggestion to consult a qualified lawyer or legal-aid service for real cases.
- **Accuracy and hallucination risk:** grounding, citation checking, and refusal behaviour reduce, but do not remove, the risk of errors; the limitations will be stated openly and evaluation results reported honestly.
- **Outdated law:** the knowledge base is a dated snapshot; amendments after that date may not be included, and the date is displayed.
- **Privacy:** no personal data is required; questions are stored without identifiers; users are advised not to enter names or sensitive details.
- **Bias and language fairness:** performance may be weaker for dialectal or very informal Bangla; the test set will include varied phrasing to measure this.
- **Data source and attribution:** official texts will be used with proper attribution and in line with the source's terms of use; the project is non-commercial and educational.
- **Sensitive topics:** for topics such as domestic violence, the answer will also show the relevant support-service information in a neutral and careful way.

5.4 Future Work

- Add more Acts, rules, and later, court judgments.
- Voice input and output for users with low reading ability.
- Automatic detection of amendments and repealed sections.
- Fine-tuning a Bangla embedding model on legal question-section pairs.
- Lawyer-reviewed answer sets and a community feedback loop.
- A mobile application and offline support.